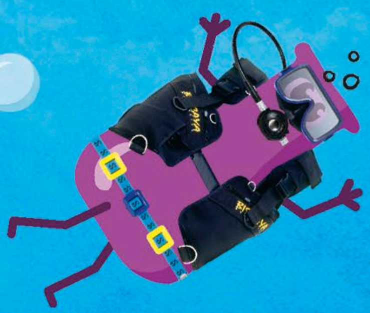
Floating

A huge metal ship can float because it's full of air. The amount of space the ship takes up weighs less than the equivalent, or same, amount of water.

Water pushes against the weight of the ship. The upward force of the water is greater than the ship's weight, so the ship floats.



Sinking

Objects sink if their weight is greater than the force of the water pushing them upwards. Dense materials, such as metal and stone, usually sink, unless they have air inside them.